

WED- May-21) Example problems

1) Consider the following velocity components.

$$u = \frac{cy}{x^2-y^2} \quad v = \frac{cx}{x^2-y^2} \quad (c: \text{cte})$$

a) prove this flow is physical- (Incomp. flow)

b) Derive streamlines of the flow.

c) If $c = 0.25$, calculate velocity at $(x,y) = (4,1)$

Solution:

a) $\nabla \cdot \vec{V} = 0$ $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \rightarrow \frac{\partial}{\partial x} \left(\frac{cy}{x^2-y^2} \right) + \frac{\partial}{\partial y} \left(\frac{cx}{x^2-y^2} \right) = ?$

$$\frac{-2cxy}{(x^2-y^2)^2} + \frac{2cxy}{(x^2-y^2)^2} = 0 \quad \checkmark$$

b) streamline definition: $u = \frac{\partial \Psi}{\partial y} = \frac{cy}{x^2-y^2}$

$$\Rightarrow \Psi = \int \frac{cy}{x^2-y^2} dy \stackrel{-\frac{1}{2}dA}{=} c \int \frac{-dA}{2A} = -\frac{c}{2} \ln |A| = \frac{-c}{2} \ln |x^2-y^2| + k(x)$$

$A = x^2 - y^2$

$$dA = -2y dy \Rightarrow y dy = -\frac{1}{2} dA$$

$$\Psi = -\frac{c}{2} \ln |x^2-y^2| + k(x)$$

(take partial) we know that $v = -\frac{\partial \Psi}{\partial x}$ and we have $v = \frac{cx}{x^2-y^2}$

$$-\frac{\partial \Psi}{\partial x} = \left[\frac{c}{2} \cdot \frac{2x}{x^2-y^2} + \frac{dk(x)}{dx} \right] = \frac{cx}{x^2-y^2} \Rightarrow \frac{dk(x)}{dx} = 0 \Rightarrow k = C_2(x)$$

(P-1)

$$\Rightarrow \boxed{\Psi = \frac{-c}{2} \ln |x^2 - y^2| + c_2}$$

$$c) \Rightarrow \begin{cases} c = 0.25 \\ x = 4 \\ y = 1 \end{cases} \rightarrow \vec{V} = \frac{0.25(1)}{(4^2 - 1^2)} \hat{i} + \frac{0.25(4)}{(4^2 - 1^2)} \hat{j}$$

$$= 0.0167 \hat{i} + 0.067 \hat{j}$$

$$|\vec{V}| = \sqrt{0.0167^2 + 0.067^2} = \boxed{0.069}$$

2) $\Psi = 4xy$

a) find velocity field.

b) prove flow is physical and irrotational.

c) find potential function

Solution:

$$u = \frac{\partial \Psi}{\partial y} \quad v = -\frac{\partial \Psi}{\partial x}$$

a) $u = 4x \quad v = -4y \rightarrow \vec{V} = 4x\hat{i} - 4y\hat{j}$

b)-1 $\vec{\nabla} \cdot \vec{V} = 0 \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 4 - 4 = 0 \quad \checkmark$

b)-2 $\vec{\nabla} \times \vec{V} = ? \quad \vec{\nabla}_x \vec{V} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 - 0 = \boxed{0}$

c) $\begin{cases} u = \frac{\partial \phi}{\partial x} = 4x \rightarrow \phi = \int 4x dx = 2x^2 + k(y) \\ v = \frac{\partial \phi}{\partial y} = \frac{dk(y)}{dy} = -4y \Rightarrow k = -2y^2 + k \end{cases}$

$$\Rightarrow \boxed{\phi = 2x^2 - 2y^2 + k}$$

③ $\Phi = A \ln r + Br \cos \theta$ $\Phi = \Phi(r, \theta)$ polar coord.

find streamfunction (Ψ)

Solution: $V_r = \frac{1}{r} \frac{\partial \Psi}{\partial \theta} = \frac{\partial \Phi}{\partial r} = \frac{A}{r} + B \cos \theta$

$\Rightarrow \left[\frac{1}{r} \frac{\partial \Psi}{\partial \theta} = \frac{A}{r} + B \cos \theta \right] \Rightarrow \frac{\partial \Psi}{\partial \theta} = A + Br \cos \theta$

$\int d\theta \Rightarrow \int \frac{\partial \Psi}{\partial \theta} d\theta = \int [A + Br \cos \theta] d\theta + f_1(r)$

$$\Psi(r, \theta) = A\theta + Br \sin \theta + f_1(r)$$

$V_\theta = \frac{1}{r} \frac{\partial \Phi}{\partial \theta} = - \frac{\partial \Psi}{\partial r}$

$\frac{1}{r} [-Br \sin \theta] = -B \sin \theta - \frac{df_1(r)}{dr}$

$\Rightarrow \frac{df_1(r)}{dr} = 0 \Rightarrow f_1(r) = k$

$\Rightarrow \Psi = A\theta + Br \sin \theta + k$

③

$$(4) \quad \phi = \frac{-A \cos \theta}{2\pi r} \longrightarrow \phi(r, \theta)$$

a) Is the flow physical?

b) find γ

Solution: write $\nabla \cdot \vec{V} = 0$ in polar and check!

$$\frac{1}{r} \left(\frac{\partial}{\partial r} (r V_r) \right) + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} = 0$$

$$\Rightarrow \frac{V_r}{r} + \frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} = 0 \quad (1)$$

$$\begin{aligned} V_r &= \frac{\partial \phi}{\partial r} = \frac{A \cos \theta}{2\pi r^2} \\ V_\theta &= \frac{1}{r} \frac{\partial \phi}{\partial \theta} = \frac{A \sin \theta}{2\pi r^2} \end{aligned}$$

Put V_r & V_θ in (1)

$$\underbrace{\frac{A \cos \theta}{2\pi r^3}}_{+1} + \underbrace{\frac{A \cos \theta}{2\pi} \left(\frac{-2}{r^3} \right)}_{-2} + \underbrace{\frac{1}{r} \frac{A \cos \theta}{2\pi r^2}}_{+1} = 0 \quad \checkmark$$

$$b) \quad \gamma = ? \quad \begin{cases} -\frac{\partial \gamma}{\partial r} = V_\theta \\ \frac{1}{r} \frac{\partial \gamma}{\partial \theta} = V_r \end{cases} \quad \checkmark$$

$$\frac{1}{r} \frac{\partial \gamma}{\partial \theta} = V_r = \frac{A \cos \theta}{2\pi r^2} \Rightarrow \gamma = \int \frac{A \cos \theta}{2\pi r} d\theta + f(r)$$

$$\Rightarrow \gamma = \frac{A}{2\pi r} \sin \theta + f(r)$$

(P-4)

$$-\frac{\partial \psi}{\partial r} = V_\theta$$

$$-\left[\frac{-A}{2\pi r^2} \sin\theta + \frac{df(r)}{dr} \right] = \frac{A \sin\theta}{2\pi r^2} \implies \frac{df(r)}{dr} = 0$$

$$\implies f(r) = \text{cte}$$

$$\implies \psi = \frac{A}{2\pi r} \sin\theta + k$$

5) $\psi = -x + y$

a) Is it physical?

b) " " irrotational? If yes, find ϕ

c) find pressure gradient

Solution:

a) $\nabla \cdot \vec{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 + 0 = 0$ satisfied!

$$\psi = -x + y$$

$$\begin{cases} u = \frac{\partial \psi}{\partial y} = 1 \rightarrow \frac{\partial u}{\partial x} = 0 \\ v = -\frac{\partial \psi}{\partial x} = 1 \rightarrow \frac{\partial v}{\partial y} = 0 \end{cases}$$

b) $\vec{\nabla}_x \vec{v} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 - 0 = 0$ ✓ irrotational ✓

$$u = \frac{\partial \phi}{\partial x} \rightarrow \frac{\partial \phi}{\partial x} = 1 \rightarrow \phi = x + f(y)$$

$$v = \frac{\partial \phi}{\partial y} = \frac{df(y)}{dy} = 1 \rightarrow f(y) = y + k \implies \boxed{\phi = x + y + k}$$

(P-5)

$$c) \frac{\partial p}{\partial x} = ?$$

use Mom. Eq.:

$$\text{integral form } \frac{\partial}{\partial t} \iiint \rho V d(\text{vol}) + \iint \rho (\mathbf{V} \cdot \mathbf{n}) V = - \iint p ds + \dots$$

$$\text{flow is steady} \Rightarrow \iint \rho (\mathbf{V} \cdot \mathbf{n}) V = - \iint p ds$$

$$\text{in 3D integral} \Rightarrow \iiint \nabla \cdot (\rho \mathbf{V} u) d(\text{vol}) = - \iiint \frac{\partial p}{\partial x} d(\text{vol})$$

for x-dir

$$\Rightarrow \text{In diff form} \Rightarrow \boxed{-\frac{\partial p}{\partial x} = \rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right)}$$

$$\begin{array}{l} u=1 \\ v=1 \end{array} \quad \begin{array}{l} \frac{\partial u}{\partial x} = 0 \\ \frac{\partial v u}{\partial y} = 0 \end{array} \quad \rightarrow \text{R.H.S} = 0 \Rightarrow \boxed{\frac{\partial p}{\partial x} = 0}$$

(P-6)

⑥ find the integral along path $(\vec{s})$ between $(0,0)$ and $(1,2)$ for the components of $\vec{V}$ in the following three cases.

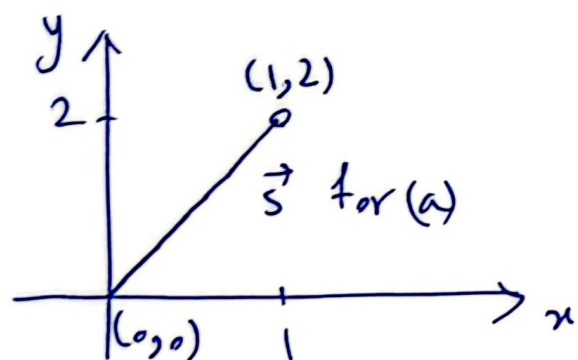
a) $\vec{s}$ is a straight line

b) $\vec{s}$ is a parabola

c) $\vec{s}$ is a portion of the x ~~axis~~ and a straight line perpendicular to it.

$$\vec{V} \text{ is } \begin{cases} u = x^2 + y^2 \\ v = 2xy^2 \end{cases}$$

Solution:



$$\int \vec{V} \cdot d\vec{s}$$

$\vec{s}$ line function is $y = 2x$

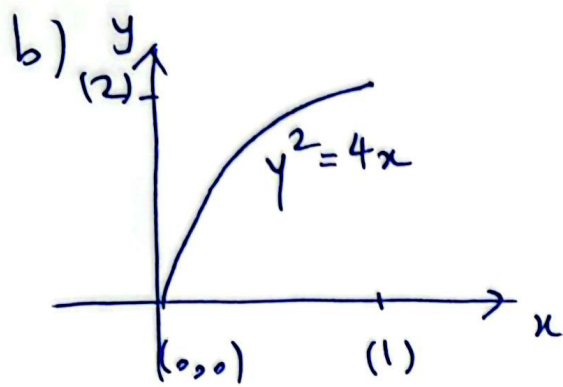
$$d\vec{s} = dx \hat{i} + dy \hat{j}$$

$$\int \vec{V} \cdot d\vec{s} = \int (u \hat{i} + v \hat{j}) \cdot (dx \hat{i} + dy \hat{j}) = \int u dx + \int v dy$$

$$= \int (x^2 + y^2) dx + \int 2xy^2 dy = \int_0^1 (x^2 + 4x^2) dx + \int_0^2 2 \frac{y}{2} y^2 dy$$

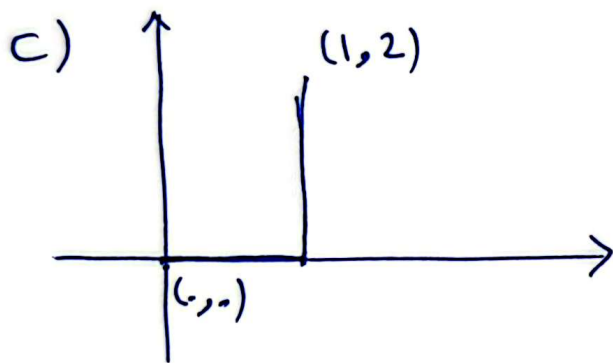
$$= \frac{17}{3}$$

(P-71)



$$\begin{aligned} \int V \cdot ds &= \int u dx + \int v dy \\ &= \int (x^2 + y^2) dx + \int 2xy^2 dy \\ &\text{over line } y^2 = 4x \end{aligned}$$

$$\Rightarrow \int_0^1 (x^2 + 4x) dx + \int 2 \left(\frac{y^2}{4} \right) y dy = \frac{83}{15}$$



along --- $\begin{cases} x=x \\ y=0 \end{cases}$

along $|$ $\begin{cases} x=1 \\ y=y \end{cases}$

$$\int V \cdot ds = \int (x^2 + y^2) dx + \int 2xy^2 dy$$

along --- $\xRightarrow{y=0} \int_0^1 x^2 dx + 0 = \frac{x^3}{3} = \frac{1}{3}$

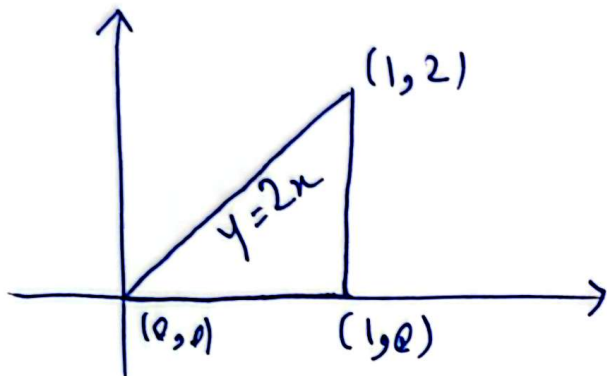
along $|$ $\xRightarrow[\frac{dx=0}{x=1}]{} \int (1+y^2) x^0 + \int_0^2 2y^2 dy = \frac{16}{3}$

$$\Rightarrow \text{---} + | = \int = \frac{17}{3}$$

(P-8)

(P-7) $\vec{V} = (x^2 + y^2)\hat{i} + 2xy^2\hat{j}$

compute circulation over the below triangle.



$$\Gamma = - \oint \vec{V} \cdot d\vec{s} = - \iint (\vec{\nabla} \times \vec{V}) \cdot \hat{n} dA$$

Solution :

$$\Gamma = - \iint (\vec{\nabla} \times \vec{V}) \cdot \hat{n} dA$$

$$\vec{\nabla} \times \vec{V} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = (2y^2 - 2y)\hat{k}$$

$$\iint (2y^2 - 2y)\hat{k} \cdot \hat{k} dx dy =$$

$$\int_0^1 dx \int_0^{2x} (2y^2 - 2y) dy$$

$$\bullet < x < 1$$

$$\bullet < y < 2x$$

$$\int_0^1 dx \left[\left[\frac{2y^3}{3} - y^2 \right]_0^{2x} \right] = \int_0^1 \left(\frac{16x^3}{3} - 4x^2 \right) dx$$

$$= \frac{4x^4}{3} - \frac{4x^3}{3} \Big|_0^1 = 0$$

$$\vec{\nabla} \times \vec{V} \neq 0 \longrightarrow \iint \vec{\nabla} \times \vec{V} = 0$$

(P-9)